GTM Agent

Project report

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Contents

List of Figures 4

List of Tables 5

I User Guide 6

1 User Guide — GTM Agent 7

1.1 What is GTM Agent? 7

1.2 Who is it for? 7

1.3 How it works 7

1.4 Features 8

1.4.1 Find the right tenders 8

1.4.2 See who to approach, and why 8

1.4.3 Tune the ranking to your judgement 8

1.4.4 Send an email that's ready to go 8

1.4.5 Rate a result, and it gets better 10

1.4.6 Agent mode for power users: Just say what you want 10

1.5 A typical session (walkthrough) 11

1.6 Common questions 11

II Project Management Report 12

2 Project Management Report 13

2.1 Overall Project Vision 13

2.1.1 Problem 13

2.1.2 Solution and Scope 13

2.1.3 Success criteria 14

2.2 Project Timeline 14

2.2.1 Gantt Chart 14

2.2.2 Milestones 14

2.3 System Architecture 15

2.3.1 Sub-systems 15

2.3.2 Data model (selected) 17

2.4 Method 17

2.4.1 Methodological principles 17

2.4.2 Hybrid matching method 17

2.4.3 Prompt Engineering 18

2.5 Technology Used 19

2.6 Current Progress and Future Plans 19

III User Acceptance Testing Report 21

3 User Acceptance Testing Report 22
3.1 Target Users Research 22
3.1.1 Target users 22
3.1.2 Reasons for choosing these users 22
3.1.3 Advantages and limitations of user research 22
3.2 User Testing Design 22
3.2.1 Test method 22
3.2.2 Testing tasks 23
3.2.3 Evaluation indicators 23
3.3 Analysis and Evaluation of Results 23
3.3.1 Main results 23
3.3.2 Core findings 23
3.3.3 Feedback processing 24
3.3.4 Conclusion 24

IV GenAI Safety and Reflection Report 25
4 GenAI Safety and Reflection Report 26
4.1 System boundary and evaluation lens 26
4.2 Safety of GenAI 26
4.3 Lessons learned and reflection 27
4.4 How our understanding of GenAI changed, and conclusion 28

References 29

Appendix A — Evaluation Protocol (UAT Script) 32

Acknowledgment of GenAI Usage 33

List of Figures

Figure 1 The Tenders screen 9
Figure 2 The Matches screen 9
Figure 3 A generated email draft 10
Figure 4 Agent mode 11
Figure 5 Project timeline 14
Figure 6 Milestone roadmap 15
Figure 7 System architecture 16
Figure 8 Per-participant UAT results 24
Figure 9 Mean UAT scores 24

List of Tables

Table 1 Target user groups 7

Table 2 Hybrid matching layers 17

Table 3 Technology stack 19

Table 4 Component status 19

Part I

User Guide

1. User Guide — GTM Agent

1.1. What is GTM Agent?

Winning work on big public construction, infrastructure and IT projects starts with finding the right opportunities, but the EU publishes thousands of new tenders every week, in dozens of languages. GTM Agent does that reading for you. It follows the EU's official tenders service (TED), picks out the ones that fit your business and, for each one, shows you the companies most worth approaching, explains *why* in plain words, and even writes a first outreach email you can edit and send.

In short: **GTM Agent turns thousands of public tenders into a short, ready-to-act list of sales leads, each with a draft email attached.** You can work through the steps yourself, or just tell the built-in assistant what you want.

60-second pitch video: `https://photos.app.goo.gl/fG1kagP8WnbJAEF17`

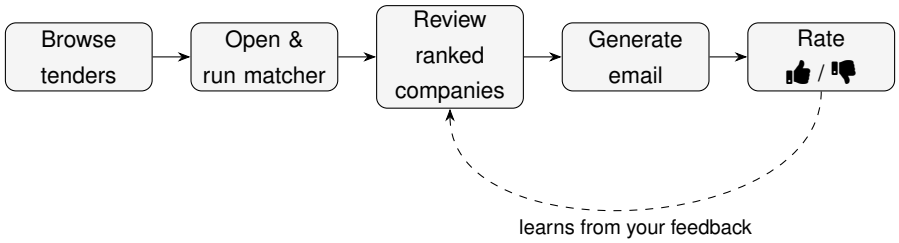
1.2. Who is it for?

Table 1 Target user groups and what GTM Agent does for each.

Role	What GTM Agent does for them
Sales / **business-development** (primary)	Finds live tenders in their field, shows which companies to approach, and hands over ready-to-send outreach emails.
Buyer / **contracting authority** (secondary)	Gets a short list of the strongest suppliers for a tender.
Power user	Runs the whole thing just by describing the goal in plain words.

1.3. How it works

Five steps take you from a tender to a ready-to-send email, or you can hand the whole thing to the built-in agent:



Browse tenders in your sectors, open one to run the matcher, review the ranked companies (each with an explainable score), generate an outreach email or a buyer digest, then rate the result 👍 / 👎, and the system learns from your feedback.

Agent mode (shortcut): prefer to just *say what you want*? The built-in agent does steps 1–4 from a single sentence and asks you to confirm if anything is ambiguous.

1.4. Features

1.4.1. Find the right tenders

The **Tenders** screen lists live EU opportunities in your field, filtered by CPV code (the EU's standard procurement categories), each showing the buyer, country, budget and deadline. Open any one to see its full details, including a link back to the original public notice, so you can always check the source for yourself (Figure 1).

1.4.2. See who to approach, and why

Open a tender and GTM Agent instantly lists the companies most likely to win it, drawn from everyone who has won a comparable EU contract in the award history it has already collected. It never hands you a mystery score: for every company you can see how well it fits on four simple measures: the kind of work it does, the procurement category it fits (its CPV code), its typical budget, and where it's based, plus a short note on what mattered most. You always know why a name is on the list (Figure 2).

1.4.3. Tune the ranking to your judgement

Know something the data doesn't, say, that a company is too small for this job? Drag a slider to give location or budget more importance, and the list re-orders as you watch. Happy with the result? Click **Apply** and GTM Agent remembers your preference for that tender, so it's still there when you come back.

1.4.4. Send an email that's ready to go

Pick a company and GTM Agent writes a tailored outreach email for you. It only ever uses real, checkable facts, the tender's reference and deadline, and the company's own track record. It even writes in the tender's own language: a French tender gets a French email. Nothing is ever sent automatically; the draft is yours to review, edit and send. You can also produce a short digest for a

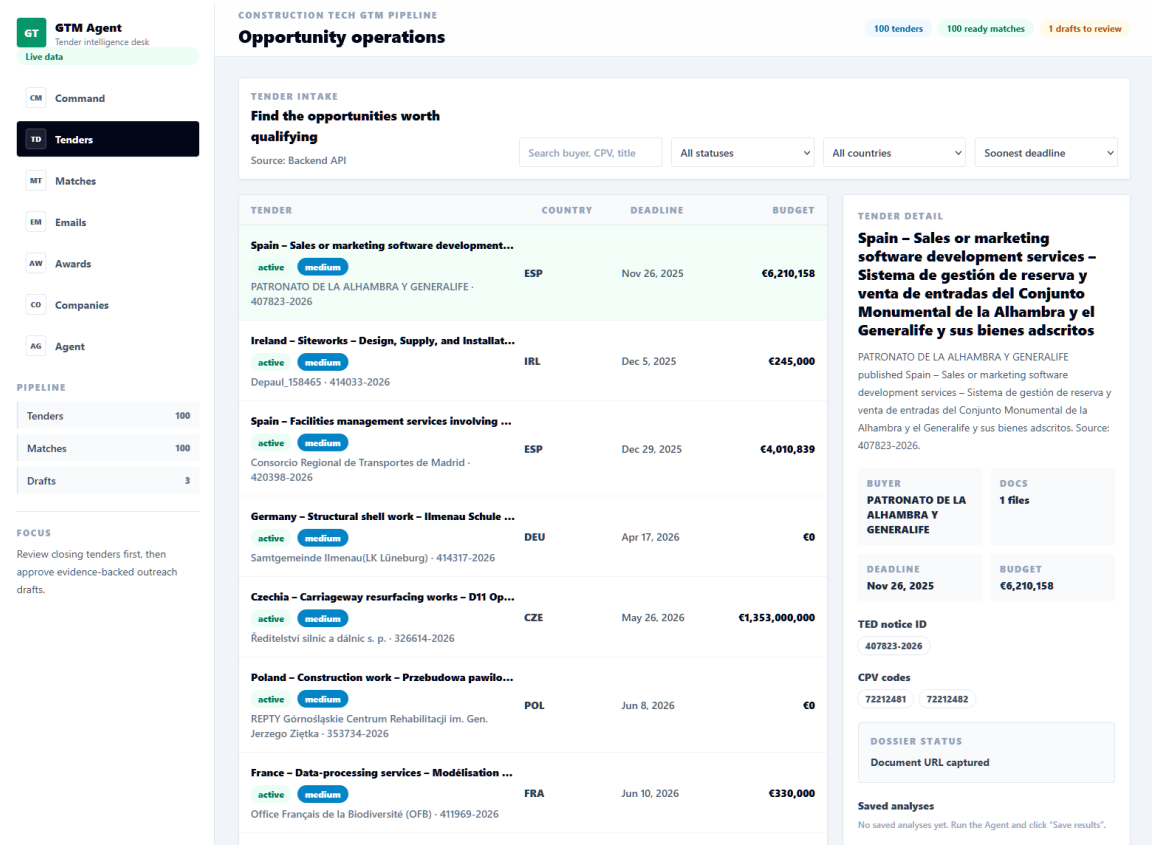


Figure 1 Browse live EU tenders on the left; open one for the full details on the right.

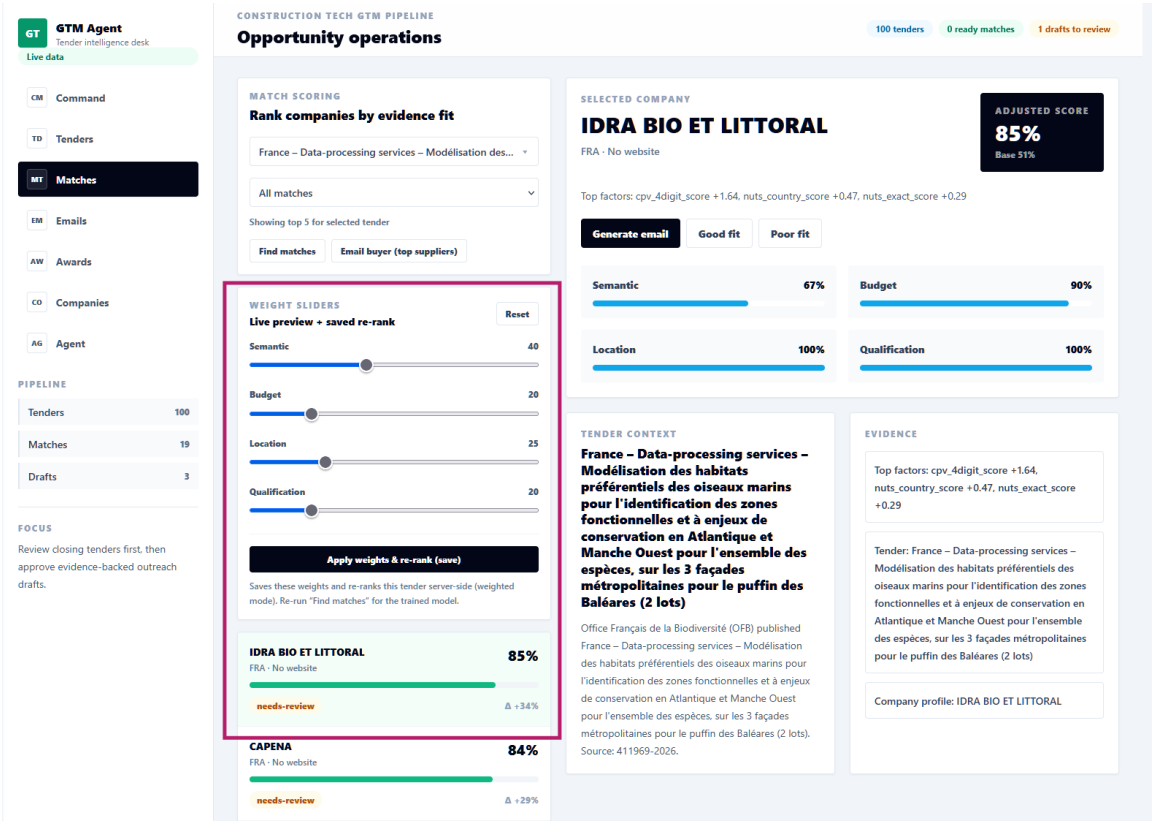


Figure 2 The Matches screen: ranked companies with a clear breakdown of why each one fits, plus sliders to re-order the list by what matters most to you.

buyer, recommending the strongest suppliers for their tender (Figure 3).

1.4.5. Rate a result, and it gets better

A quick thumbs-up 👍 or thumbs-down 👎 on any match or email tells GTM Agent what a good lead looks like to your team. The more you use it, the more its suggestions lean toward the kind of leads you actually want.

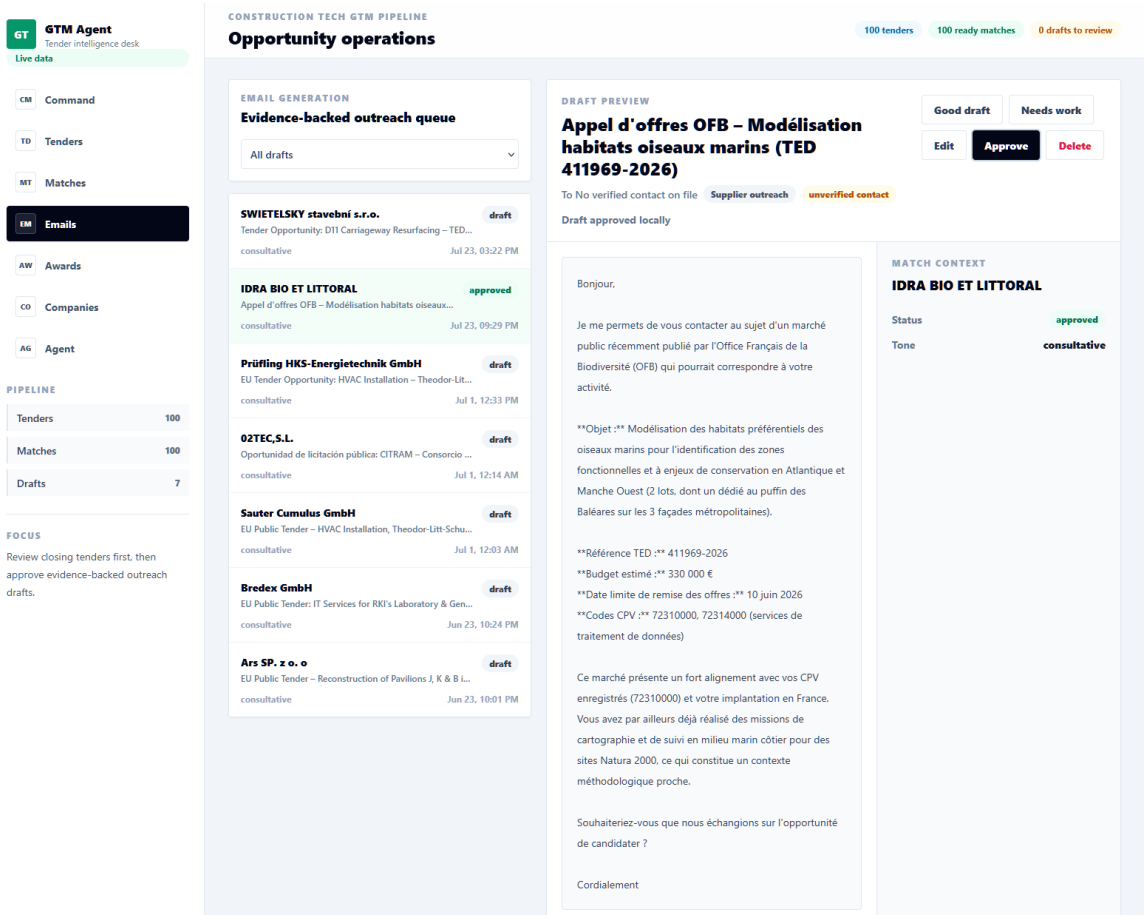


Figure 3 A ready-to-send draft, written only from real facts and in the tender's own language.

1.4.6. Agent mode for power users: Just say what you want

Prefer not to click around? Tell the built-in assistant, in plain words, for example, *"Find the tender about school renovation in Germany, extract its certification requirements, and draft an outreach email to whichever supplier best qualifies."* It works through the steps for you and shows what it's doing as it goes (Figure 4). A few things make it easy to trust:

- **Name a tender any way you like**, by its title or its official TED notice number (for example 420398-2026). If several could match, it stops and asks you to pick one instead of guessing.
- **It stays on your data.** An "Offline mode" toggle is on by default, so it only uses the tenders and companies already loaded and won't reach out to the internet.
- **One-click follow-ups.** When it's done, **save the result** or **draft an email** to any of the top matches in a single click.

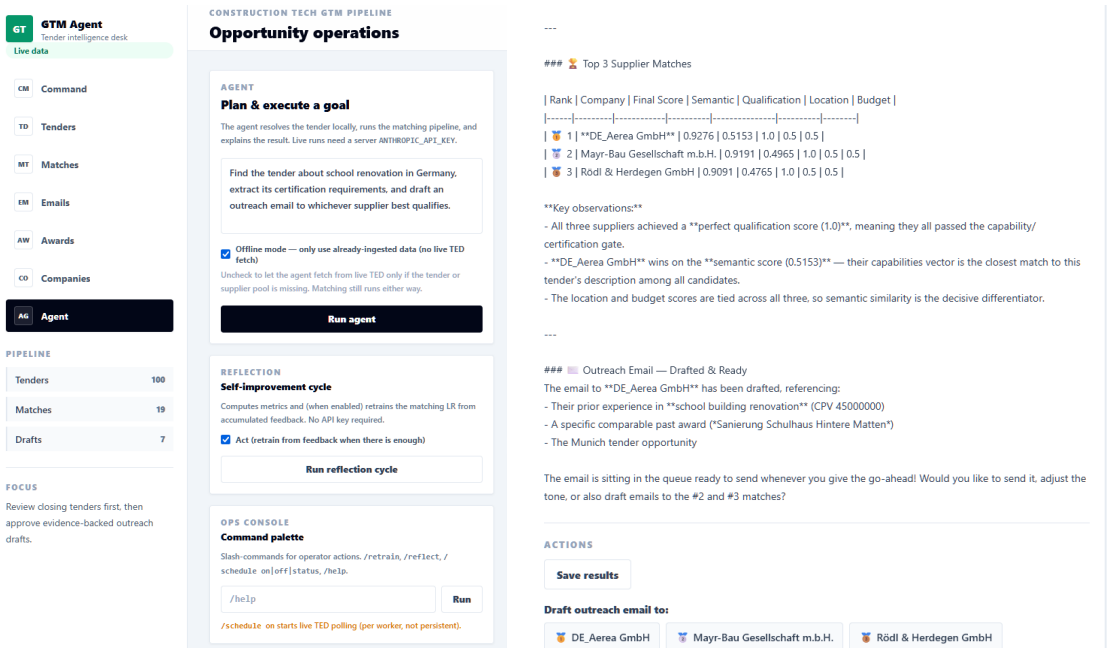


Figure 4 Ask in plain words: the assistant runs the steps live (left) and returns a ranked result (right).

1.5. A typical session (walkthrough)

1. Open the **Tenders** screen and spot *"Czechia, carriageway resurfacing works, D11."*
2. Open it and view the matches, the top results are genuine Czech road builders (EUROVIA, STRABAG, SWIETELSKY).
3. Open the top company to read why it fits and what it's based on.
4. Generate an email and review the draft, every claim traces back to a real source.
5. Edit, give it a thumbs-up, and you're done: a sourced lead and a ready email in under a minute.

1.6. Common questions

- **Does GTM Agent send emails on its own?** No, it only writes drafts for you to review. You're always in control.
- **Where does the data come from?** Only from public EU tender records, no private or personal data.
- **Can it handle other languages?** Yes, it reads tenders in any EU language, and writes each email in the tender's own language.
- **What fields does it cover?** Building/construction and IT services to start, and it can be set up for other fields.
- **Can the assistant do things on its own?** Nothing surprising. It only uses data already loaded, asks before acting when something is unclear, and never sends an email by itself.

Part II

Project Management Report

2. Project Management Report

2.1. Overall Project Vision

2.1.1. Problem

In B2B sales for construction, infrastructure and IT/services projects, business development teams spend significant time finding relevant tenders and identifying plausible bidders. Public contracts above the EU thresholds must be advertised Union-wide (Directive 2014/24/EU), which makes TED an exhaustive but very high-volume source. Because TED does not reveal active bidders during the tendering phase, teams still rely on manual portal searches, company research, and fragmented market knowledge, leading to wasted effort, missed opportunities, and missed deadlines.

2.1.2. Solution and Scope

GTM Agent turns public TED data into actionable sales leads. The MVP focuses on construction and IT/services tenders filtered by CPV codes, the EU's standard procurement classification (Regulation (EC) No 2195/2002). It combines live Contract Notices with historical Award Notices, using CPV, buyer, and budget similarity to identify companies likely to bid on similar projects again.

Key deliverables (in scope)

1. Automated ingestion of TED contract notices and award notices, plus dossier-PDF download where the document is directly reachable.
2. Entity resolution of award winners into canonical company records.
3. A hybrid tender↔company matcher with explainable sub-scores.
4. Evidence-grounded, two-sided email generation (supplier + buyer).
5. A feedback capture and weight-retraining loop.
6. A Claude tool-use agent that plans and executes the pipeline from a natural-language goal.
7. A Vue web dashboard and a documented REST API.

Boundaries

1. No credential-juggling scraping of authenticated buyer portals — restricted dossiers are flagged for manual review rather than scraped (see the Safety report).
2. No translation layer — multilingual content is handled natively by a multilingual embedding model rather than machine translation.
3. No authentication / multi-tenancy in Phase 1.
4. No automatic sending of emails — drafts are produced for human review only.
5. Vision-LLM blueprint parsing and a Neo4j incumbency knowledge graph are catalogued as **Future Work**, not committed to the MVP.

2.1.3. Success criteria

The core evaluation method of the project is backtesting. For the historical tender, the system goes back to its release time, runs the matcher only with the information available at that time, and observes the ranking of the real winning company in the results. The stage target is: Top-10 hit-rate $\geq$ 50% in the MVP stage, increase to $\geq$ 65% after mid-term, and the final target is $\geq$ 75%. Because the implemented backtest reports hit-rates only up to Hit@5, these stage targets were operationalised as Top-1 $\geq$ 25% and Top-5 $\geq$ 55%. The final model meets both (Hit@1 44.2%, Hit@5 57.6% on 500 held-out awards; full figures and their limits are discussed in Part IV); the aspirational Top-10 $\geq$ 75% was not separately measured and no claim is made about it.

2.2. Project Timeline

2.2.1. Gantt Chart

The project is organised into eleven phases over twelve teaching weeks, clustered into five workstream categories: Ingestion, Understanding & Matching, Generation & Agent, Frontend & Demo, and Documentation & Evaluation. Figure 5 shows how the workstreams overlap and where the milestones fall.

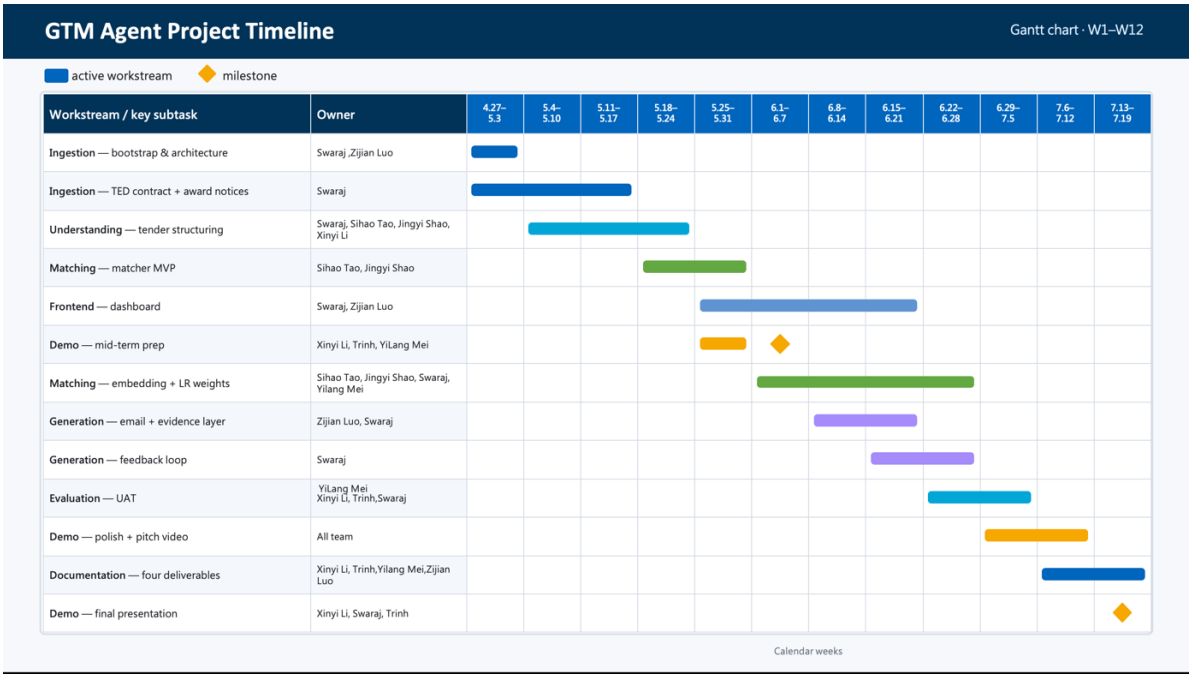


Figure 5 Project timeline (workstreams and milestones over teaching weeks W1–W12).

2.2.2. Milestones

The roadmap in Figure 6 groups the same twelve weeks into three stages. It moves from data ingestion and MVP construction to matching, email, and retraining features. The final stage covers UAT, documentation, presentation, and submission.

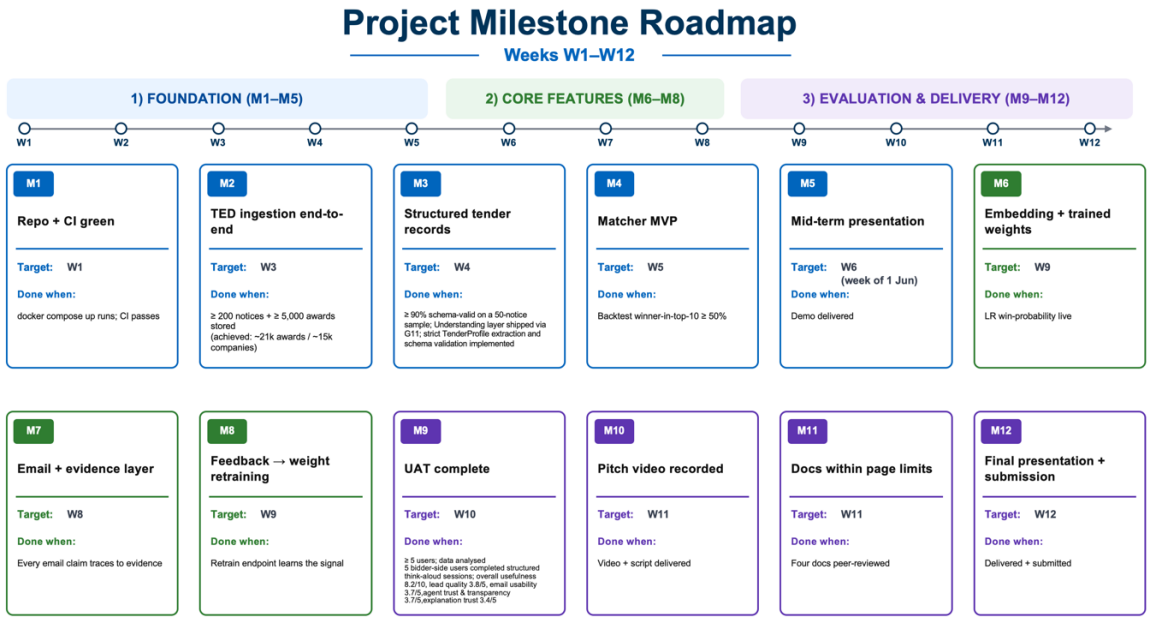


Figure 6 Milestone roadmap across the three project stages.

2.3. System Architecture

GTM Agent is organised into seven sub-systems, spanning data ingestion, storage, matching, evidence-grounded generation, and agent orchestration. Figure 7 shows how they connect, and the paragraphs below describe each in turn.

2.3.1. Sub-systems

1. Ingestion. The module retrieves live Contract Notices and historical Award Notices through the TED v3 Search API and stores them in PostgreSQL. Accessible dossier PDFs are downloaded, validated, deduplicated with SHA-256, and converted to text. Command-line jobs include retry and rate-limit handling to improve reliability.

2. Entity Resolution. The module standardizes company names by normalizing capitalization, punctuation, and legal suffixes. It matches firms primarily through reliable identifiers such as tax IDs and otherwise uses canonical names, while filtering invalid identifiers to prevent incorrect merges.

3. Tender Understanding. The module converts notice and dossier content into a structured TenderProfile, covering fields such as budget, deadline, CPV codes, qualifications, and technical requirements. Outputs follow a strict JSON schema and are validated through CPV repair, budget checks, and deduplication.

4. Matching. The matching module builds company profiles from historical awards, including capabilities, embeddings, project scale, and location. It then filters candidates by CPV, region, budget, and qualifications before using semantic features and a Logistic Regression model whose output is

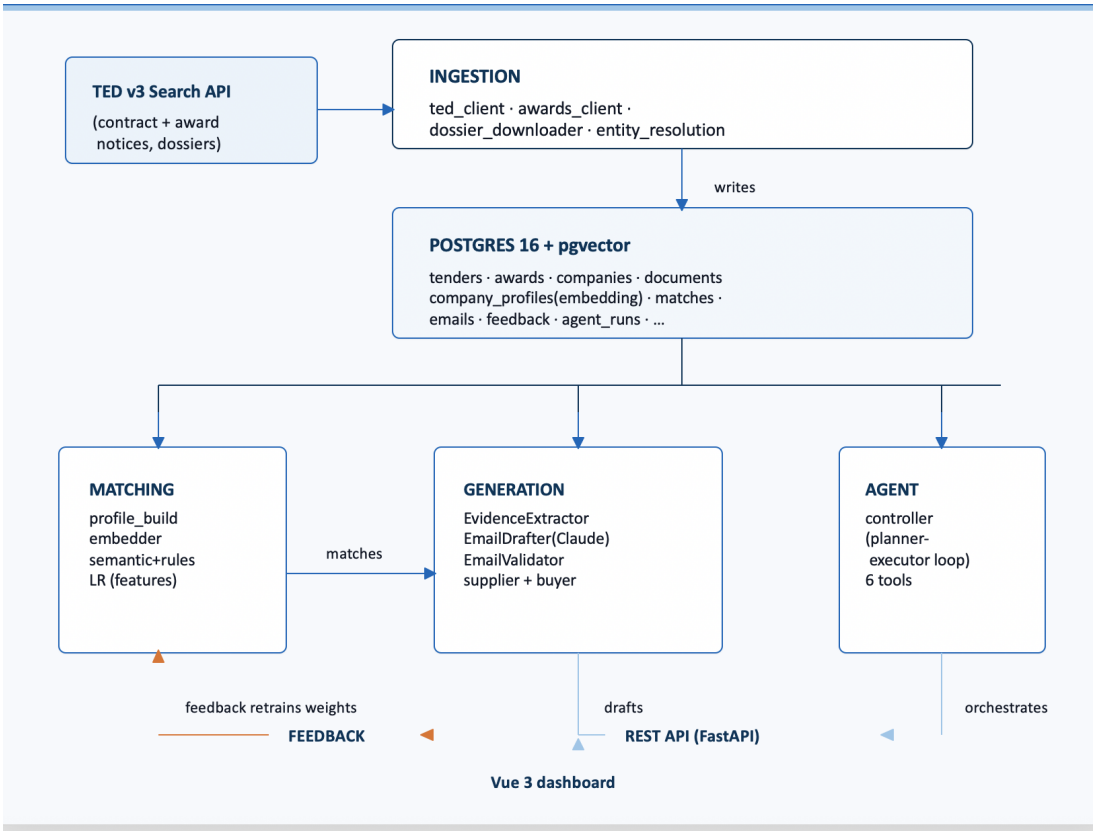


Figure 7 System architecture: ingestion, storage, matching, generation, and agent.

calibrated to a usable win probability (Niculescu-Mizil & Caruana, 2005), implemented in scikit-learn (Pedregosa et al., 2011). Sub-scores and explanations show users why each company is ranked.

5. Generation. The system generates supplier outreach emails and buyer tender digests. An Evidence Extractor supplies verified facts, and Claude drafts the email under evidence constraints, with a deterministic template as fallback. An Email Validator rejects unsupported claims, and all drafts require user review before sending.

6. Agent. The Agent uses Claude's tool-use capability to turn natural-language goals into executable steps, including data retrieval, matching, email drafting, and read-only database queries. A planner–executor workflow streams each step to the interface. For safety, write tools default to dry-run mode, database access is limited to read-only SELECT queries, and external requests are strictly bounded.

7. API and frontend. The FastAPI backend provides REST endpoints for tenders, awards, companies, matches, emails, feedback, and Agent functions, with automatic OpenAPI documentation. The Vue 3 frontend supports tender browsing, candidate review, weight adjustment, score explanations, and email approval. Sample data is used as a fallback when the backend is unavailable.

2.3.2. Data model (selected)

The system stores tenders, awards, companies, profiles, matches, emails, feedback, and Agent runs in a unified PostgreSQL schema. `company_profiles` uses pgvector to store 384-dimensional embeddings for similarity matching, while a general embeddings table supports future models. Company master data and capability profiles are kept separately because they have different sources and update cycles. Full schema details are documented in `docs/Database_Explanation.md`.

2.4. Method

2.4.1. Methodological principles

Three principles shaped every technical decision:

1. **Classical ML over LLM where one exists.** The ranking of the matches is done by embeddings and logistic regression, not by asking an LLM, giving reproducible and explainable scores.
2. **Open-source, locally-run embedding models.** The most frequent operation in the pipeline — semantic similarity between a tender's requirements and a company's portfolio — runs locally, so there is no per-query API cost.
3. **Constrain the LLM to extraction and drafting.** The LLM never does free-form ranking or makes unverified factual claims. Because fluent but unsupported output is a known failure mode of generative models (Ji et al., 2023), restricting the model to evidence-backed information is a design constraint rather than a stylistic preference.

2.4.2. Hybrid matching method

The core engineering of GTM Agent is its hybrid matching pipeline, which decomposes the problem into independent layers, each using the right tool (Table 2).

Table 2 Hybrid matching: each layer uses the right tool.

Layer	Tool
Structured field extraction	LLM (Claude)
Semantic similarity	Open-source sentence embeddings
Budget / location / qualification	Deterministic rules
Weight fusion	Logistic regression

1) Hard disqualification. Claude extracts structured fields from the tender and company sources but does not rank yet. Before any scoring, a disqualifier removes clearly ineligible candidates using deterministic filters (blacklisting, geographic eligibility, company size versus budget tier, or tender-specific exclusions), so the more expensive scoring steps only run on cleaner shortlists.

2) Semantic similarity. For each surviving candidate, a sentence embedding model converts each text into a fixed-length vector, positioning semantically related texts close together and making similarity computable as a cosine distance rather than as word overlap (Reimers & Gurevych, 2019).

The model used is a distilled MiniLM (Wang et al., 2020) whose multilingual variant is trained so that a sentence and its translation land at the same point in the vector space (Reimers & Gurevych, 2020). That property is what lets a Czech tender be matched against a German company profile without a translation step — necessary for a platform like TED.

3) Rule-based checks. Budget overlap, geographic fit, and CPV qualification are covered by deterministic sub-scores, because they involve structured data with unambiguous answers: budget fit (after FX normalisation to EUR), geographic fit (NUTS regions, Regulation (EC) No 1059/2003), and hierarchical CPV qualification (an exact 8-digit CPV scores 1.0).

4) Logistic-regression weight fusion. A logistic regression, trained on historical award notices with the ground truth "did this company actually win?", combines ten features into a calibrated win probability; its coefficients give a clear per-feature explanation. A feedback loop retrains it on thumbs-up/thumbs-down pairs, storing a lasting feature snapshot with each feedback row so re-matching never destroys training data.

2.4.3. Prompt Engineering

GenAI is used in three places, each with a deliberate prompting strategy.

Structured field extraction. A chain-of-thought-style prompt, following the step-by-step decomposition principle of Wei et al. (2022), asks the model to describe the type of section it is reading, list candidate fields, and produce a JSON fragment following a strict schema. Only the JSON is processed and checked; shape failures are rejected. The model is prompted in English while foreign-language source text is included verbatim, using its multilingual reading ability without a translation step.

Evidence-grounded email drafting. A two-step pattern: a deterministic extractor first builds the evidence list (TED ID, CPV codes, deadline, match scores, company facts); the model then receives only that evidence and is instructed to add no claim outside it. Output is post-validated, and any unsupported entity or claim flags the draft for review rather than sending. The prompt also tells the model to match the tender's language automatically.

Agent orchestration. The agent follows the reason-then-act pattern of Yao et al. (2023): it alternates a short planning step with a tool call and feeds the tool result back before deciding the next step. A planner–executor system prompt defines the goal, the tool set (as JSON schemas), and safety constraints (DB-first resolution, offline-by-default ingest, read-only SQL, `ask_user` for ambiguity). Tool definitions and the system prompt stay stable across turns to benefit from prompt caching, and messages persist to `agent_sessions` so a paused run resumes exactly where it left off.

2.5. Technology Used

The stack is a Python/FastAPI backend, a PostgreSQL + pgvector database, and a Vue 3 frontend, with open-source models for ranking and Claude for generation and the agent (Table 3).

Table 3 Technology stack.

Layer	Technology	Notes
Language / runtime	Python 3.11	Pinned for dependency compatibility
HTTP framework	FastAPI 0.115	First-class Pydantic v2 + OpenAPI
Validation	Pydantic v2	Strict schema enforcement
ORM / migrations	SQLAlchemy 2.0 / Alembic	Typed `Mapped[]` API
Database	PostgreSQL 16 + pgvector	Single schema; vector similarity in-DB
HTTP client	httpx (+ truststore)	`requests` banned by spec
PDF text	PyMuPDF	Dossier extraction
Embeddings	sentence-transformers: paraphrase-multilingual-MiniLM-L12-v2 (384-d), CPU-only	Multilingual MiniLM; lazy-loaded
Classical ML	scikit-learn + joblib	Trained logistic-regression matcher (calibrated win-probability)
LLM	Anthropic Claude (`claude-sonnet-4-6`)	Agent, reflection, email drafting
Scheduler	APScheduler (in-process)	Opt-in cron loop (ingest + reflect); runtime on/off/status toggle
Frontend	Vue 3 / TypeScript / Vite 5 / Pinia / Tailwind	Three-column dashboard
CI	GitHub Actions	pytest · lint+build · migration smoke
Containerisation	Docker Compose	db + backend + frontend

2.6. Current Progress and Future Plans

Every planned MVP component is implemented, and the components on the critical path have been smoke-tested against live TED data; Table 4 summarises their status, followed by known limitations and future work.

Table 4 Component status.

Component	Status
Repo scaffold, schema, CI	Done
TED contract-notice ingestion	Done: CPV-filtered, live-verified

Component	Status
Award back-fill + entity resolution	Done: ~21k awards / ~15k companies cleaned and stored; 24-month back-fill planned
Dossier downloader + ingestion CLI	Done (scoped): ingestion-only utility; thin coverage
Understanding layer (Claude field extractor)	Done: tender profile persisted; soft fields only
Matching (hybrid + trained model)	Done: multilingual embeddings (MiniLM), rules, and a calibrated logistic regression; returns top-N with sub-scores; live-smoked
Ranking + explanations	Done: per-supplier sub-scores plus explanation; numbers only from `query_db`, live score recalculation runs in the browser with no round-trip
Email generation (Claude)	Done: constrained generation only; all evidence validated and grounded; live-verified
Feedback capture + retraining	Done: feedback retrains the model
Scheduler (opt-in) + reflection loop	Done: reflections stored; runtime on/off/status control
Agent (DB-first, streaming, resumable, result actions)	Done: planner–executor loop, max 20 turns, every step streamed live; pauses for user input and resumes from the saved session; live-smoked
Ops command console	Done: `/retrain`, `/reflect`, `/schedule`, `/help`
REST API (7 routers) + Vue dashboard (7 views)	Done: browse tenders, open detail pages, inspect recommendations, approve and send email; sample-data fallback
ML training dataset export + retrain on 80% of awards	Done: 16,812 training awards

Limitations and future plans. Company profiles come from award history only, so the supplier pool is limited to companies that have won a listed EU tender; per-portal dossier scraping is prepared but not yet built, and the external match fallback (suggesting and scraping websites when no strong internal match exists) is out of scope. This fixed supplier pool also explains why the certification disqualifier, although built and linked to the extraction layer, stays inert in production: the award data carries no supplier-certification field. Because TED publishes no supplier email addresses, supplier recipients are website-derived and marked unverified rather than fabricated, consistent with the rule never to create information it cannot trace to a source.

Three UAT interface requests remain future work at the presentation layer (a city field in the detail view, a company-website link in the email and detail views, and shorter tender titles); they were deferred to focus on the agent work. Match alerts and competitor analysis are also valuable requests that reuse existing components — alerts as a scheduled query over stored matches, competitor analysis as aggregation over already-ingested award history. Finally, vision-model blueprint parsing, a Neo4j incumbency knowledge graph, and an embedding ensemble (adding `all-mpnet-base-v2` and `multilingual-e5-large`) are catalogued but not committed.

Part III

User Acceptance Testing Report

3. User Acceptance Testing Report

3.1. Target Users Research

3.1.1. Target users

This UAT invited a total of five bidder-side users from the transportation, energy, and construction industries. Their positions include Tender Specialist, Project Development, Sales Support, Business Development, and company CEO. The first four are direct daily users of the system, while the CEO represents the manager who ultimately decides whether the enterprise will participate in a given tender.

3.1.2. Reasons for choosing these users

These users are directly related to the core usage scenarios of GTM Agent and can judge, from an actual business perspective:

1. whether the company recommended by the system is a reasonable potential bidder;
2. whether the matching score and explanation are credible enough;
3. whether the weight adjustment is in line with business judgment;
4. whether the operation process of the Agent is transparent and controllable.

3.1.3. Advantages and limitations of user research

Testing with real industry personnel yields direct feedback with business context and checks whether the product really solves the problem of bidding discovery and potential-company screening. The participants' positions are diverse, including both daily executives and decision-makers. The limitation is that the sample size is only five people, so no general statistical inference can be made; participants' familiarity with the bidding process also varies. Five participants are a common working minimum for qualitative usability testing (Nielsen & Landauer, 1993), but that argument concerns the discovery of interface problems, not the estimation of population means: the rating averages below are indications, not measurements. In addition, the test uses system data and scenarios prepared by the team, not the real private data of the participants' enterprises.

3.2. User Testing Design

3.2.1. Test method

Each participant completes a structured think-aloud session (Ericsson & Simon, 1993), verbally expressing their judgments, confusions, and expectations while working, so that reasoning is captured as it happens rather than reconstructed afterwards. The test combines task scenarios with a post-test questionnaire: first observing whether the user can complete the core process, then collecting feedback through scale scoring and open-ended questions. The UAT script is in Appendix A.

3.2.2. Testing tasks

Participants complete three core tasks in turn:

1. **Looking for candidate companies.** Choose a tender, check the companies recommended by the matcher, and judge whether the top-ranked company is a credible potential bidder.
2. **Adjust the matching weight.** Use the weight sliders to adjust factors such as semantic matching, budget, region, and qualification, and observe whether the ranking changes in real time as expected.
3. **Generate and review emails.** Generate an evidence-based outreach email for a candidate company, check whether it relates to the tender and company background, and judge whether it is usable after minor edits.

3.2.3. Evaluation indicators

The test mainly evaluates: task completion, task time, lead quality (are the recommended companies reasonable), explanation trust (does the user understand and trust the ranking), email usability (usable after minor edits), agent trust and transparency, overall usefulness / recommendation intention, and open feedback (obstacles encountered and desired features). Scale items use 1–5 Likert points (Likert, 1932), and the overall recommendation item uses the 0–10 Net Promoter question (Reichheld, 2003).

3.3. Analysis and Evaluation of Results

3.3.1. Main results

The five participants rated overall usefulness at 4.1/5 in the post-test questionnaire and willingness to recommend at 8.2/10; lead quality at 3.8/5, with email usability and agent trust both at 3.7/5. Explanation trust had the lowest score, 3.4/5. Every participant's recommendation intention was at least 7/10, indicating overall recognition of the system's value. Scored as a Net Promoter Score (two promoters, three passives, no detractors) the result is $+40$, although with five participants a single detractor would move that by twenty points. All five reached the end of the core workflow: two completed it fully and independently, while the other three needed minor facilitator guidance and are recorded as partial completions. Time on task ranged from 18 to 28 minutes, with the fastest completions by the most frequent TED users. Figures 8 and 9 report the per-participant results and the resulting means.

3.3.2. Core findings

Users generally believe the system reduces the time spent manually browsing TED and searching for potential companies, and that the recommendations carry a certain business rationality. However, users' understanding of "why the ranking changes" is weak, so the natural-language explanation still needs strengthening — reflected in the lowest mean score, explanation trust (3.4/5). Participants could nonetheless read the named ranking factors (semantic match, budget, region, qualification). All participants also asked that the email directly present the tender's core information, such as

	1) Tender Specialist - Transportation BIDDER-SIDE FOCUS Daily User	2) Project Development - Energy BIDDER-SIDE FOCUS Direct target user	3) Sales Support - Construction BIDDER-SIDE FOCUS Direct target user	4) Business Dev. - Construction BIDDER-SIDE FOCUS Direct target user	5) CEO - Construction BIDDER-SIDE (Exec.) FOCUS Approves tenders pursued
Task success	Partial / Yes	Yes	Partial	Yes	Partial
Time on task	25 min	18 min	24 min	20 min	28 min
Lead quality	4/5	4/5	3.5/5	4.5/5	3/5
Explanation trust	3.5/5	4/5	3/5	3.5/5	3/5
Email usability	3.5/5	4/5	3/5	4/5	4/5
Agent trust & transparency	4/5	4/5	3/5	4/5	3.5/5
Overall usefulness / NPS	10/10	8/10	7/10	9/10	7/10
Overall impression	Very **positive**; strong potential; needs richer info	**Positive**; useful for faster discovery	**Mixed-positive**; needs clearer explanations.	Very **positive**; strong value for competitor analysis	**Positive**; helpful but needs more guidance
	4.0 / 5	**4.0 / 5**	**3.2 / 5**	**4.1 / 5**	**3.4 / 5**

Figure 8 Per-participant results; scale items use 1–5, the recommendation item 0–10.

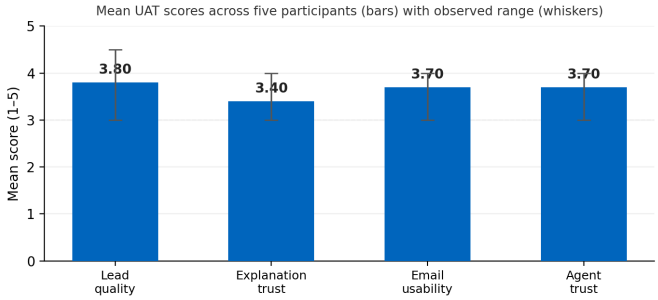


Figure 9 Mean UAT scores, whiskers showing the observed range.

budget, deadline, and location.

3.3.3. Feedback processing

In response to concerns about insufficient email information, the team added a tender summary block with budget, deadline, and location, removing the need to return to the tender page before sending. The Agent Q&A was also expanded so users can query tender, company, and matching data, not just request recommendations. Requests that could not be met within the MVP — city fields, company-website links, shorter tender titles, match alerts, and competitor analysis — were recorded as future work.

3.3.4. Conclusion

UAT shows that GTM Agent does address the practical problems of target users in discovering bidding opportunities, screening candidate companies, and preparing initial outreach. Users most valued the efficiency gain and the lead quality, but trust in the ranking explanation remains the most important area for improvement. Subsequent development should therefore prioritise the natural-language explanation, supplement key tender and company information, and retain human review of emails and results.

Part IV

GenAI Safety and Reflection Report

4. GenAI Safety and Reflection Report

Scope and evidence. GTM Agent supports EU-procurement business development by ingesting public TED notices and historical awards, ranking candidate suppliers, and drafting outreach for human review. This report draws on the implemented system, the ranking backtest, and a five-participant user-acceptance test (UAT), and keeps implemented controls, evaluation findings, and residual boundaries distinct so that design intent is not mistaken for evidence of safety.

4.1. System boundary and evaluation lens

Where GenAI is used. GTM Agent does not hand the whole decision chain to one model. GenAI performs three bounded tasks: extracting soft fields into a schema, drafting emails from a structured evidence object, and orchestrating a restricted tool set. Supplier ranking stays outside the generative path (multilingual embeddings, deterministic rules, and logistic regression), keeping the recommendation measurable and explainable while confining the model to language and orchestration.

What counts as evidence. The vision specified a two-sided MVP, a hybrid matcher, feedback and retraining, reviewable drafts, an agent, a ranking backtest, and UAT. The final model, trained on the earlier 16,812 awards and evaluated on 500 chronologically later held-out awards, reached Hit@1 / Hit@3 / Hit@5 of 44.2% / 54.0% / 57.6%, meeting the Top-1 ($\geq 25\%$) and Top-5 ($\geq 55\%$) targets. Two boundaries qualify that number. The test set is restricted to tenders whose winner is a company the system already knows, so it measures ranking, not coverage of unknown firms; but the winner is not exempted from the hard filter, and the 26.6% of cases in which the filter removes it are counted as misses rather than excluded. Hit@10 and MRR were not reported, so no claim is made about them.

4.2. Safety of GenAI

Risks arise from generated outputs, agent actions, untrusted inputs, and operation.

Factual integrity of emails. Emails are where generated content reaches users, so an inaccurate claim about a supplier, a deadline, or prior work is the primary output risk — and fluent, confident, unsupported text is a well-documented failure mode of generative models rather than an occasional defect (Ji et al., 2023). The system assembles the tender record, supplier profile, scores, match explanation, and document excerpts, then asks Claude to draft from that evidence alone; if the model call, JSON parsing, or validation fails, a deterministic template is built from the same evidence. Every output is stored as a `draft` and nothing is sent automatically. These measures limit unsupported content but do not verify every statement: the current `EmailValidator` checks subject, body, and length, not each claim against a source, so human review remains necessary, and a production version should require every claim to resolve to an identifiable source.

Agent authority. An early version treated `dry_run` as a safety switch, but it blocked database writes without blocking network calls, so a routine request could start costly TED polling and document scanning. The revised agent makes dry-run network-free, resolves tenders from the local database by default, and removes bulk dossier download entirely. Live ingestion is bounded (look-back window, row limit, once per goal, with an extra condition on award ingestion); the database tool is read-only (write verbs blocked, row cap, read-only connection); and `ask_user` pauses on ambiguity rather than guessing. The operator console (scheduler, retraining) is still unauthenticated and would need access control and an audit policy before deployment.

Untrusted inputs, provenance, and privacy. Tender and dossier text is attacker-influenced and may carry instructions unrelated to the task — the indirect prompt-injection channel described by Abdelnabi et al. (2023), in which the payload reaches the model through retrieved content rather than through the user's own turn. Schema-constrained extraction, evidence-bounded drafting, restricted tools, and downstream validation limit — though do not eliminate — that channel: untrusted text cannot alter the deterministic ranking path or gain general write access. The data is public TED and award records, not private customer data; raw payloads, the evidence object, and each document's source URL and content hash are retained; files behind buyer-portal logins are marked `restricted` rather than scraped. Two qualifications follow. Supplier contact addresses derived from company websites are still personal data under Regulation (EU) 2016/679 even though their source is public, so deployment would need a lawful basis, retention periods, role-based access, and a privacy review. And because the system generates text that is sent to people on the system's suggestion, the transparency and human-oversight duties of Regulation (EU) 2024/1689 are the relevant compliance frame for a production version, even though a lead-generation assistant is not itself a high-risk system under that regulation.

Reliability, cost, and fairness. Rate limiting and retries bound TED ingestion, local embeddings cut external calls, and a train–serve feature-contract test guards the deployed matcher, with the deterministic email fallback as an explicit degradation path. Reliability is not fairness, however: award-history training can entrench incumbents, and although component scores and adjustable weights make the basis of a recommendation visible, they do not remove that advantage. With no fairness metric yet reported, future evaluation should measure ranking quality for smaller and newer suppliers, not only the recovery of past winners; the ranked group-fairness criterion of Zehlike et al. (2017), which constrains the representation of a protected group within every prefix of a top-k list, is a concrete candidate measure for that, with firm age or size as the grouping attribute.

4.3. Lessons learned and reflection

Delivery versus evidence. The MVP integrated ingestion, entity resolution, hybrid matching, evidence-backed drafting, feedback and retraining, the agent, an API, and a dashboard, but a component existing does not by itself prove product value. The backtest gives quantitative evidence for the matcher and UAT adds workflow-level evidence; together they describe completion better than a checklist.

What the ranking KPI teaches. The matcher met its Top-1 and Top-5 targets, yet raising the training share from 50% to 80% did not improve Hit@3 (54.8% to 54.0%). The two runs use different held-out slices, so this is not a controlled comparison and the roughly one-point difference is within noise; the reading we take from it is therefore modest — more data of the same kind did not move the metric. That points to a representation limit rather than too few examples, since capability profiles are built from thin award-history text. Enriching those profiles or adding an embedding ensemble is the more promising direction, and a controlled ablation on a fixed slice would be needed to confirm it.

What the UAT teaches. Where the backtest asks whether the system retrieves the known winner, UAT asks whether users can interpret and act on the result. Five bidder-side participants from transportation, energy, and construction completed think-aloud sessions covering candidate discovery, weight adjustment, and email drafting; the full per-participant results are in Part III, Figure 8. Perceived lead quality (3.8/5) exceeded explanation trust (3.4/5): users saw the business value of the candidates but not always why each ranked where it did. Showing a score is not the same as explaining it, so a future interface should tie each recommendation to its CPV, geographic, budget, and semantic evidence. With only five participants these are design priorities, not population-level effects.

Practices that worked, and what to change. Assigning ranking to classical ML and rules kept the path measurable and auditable, and repeated smoke tests against live TED data caught defects (qualification scoring, budget normalisation) that an interpretable architecture alone would not. The main organisational lesson is to agree the interfaces between parallel workstreams early — the in-app and standalone matchers were unified late — and to treat evaluation the same way: each KPI needs a fixed dataset snapshot, an owner, a reproducible script, and a decision date from the start. Scope discipline (deferring authenticated scraping, multi-tenancy, blueprint parsing, and a knowledge graph) kept the MVP focused; the next stage should add claim-to-source email checks, authenticated operator controls, a larger UAT with a human baseline, and fairness analysis.

4.4. How our understanding of GenAI changed, and conclusion

The project reframed GenAI as a language-and-orchestration component bounded by evidence objects, schemas, permissions, and user confirmation — not a general decision-maker; the dry-run incident showed that a control matters only when it blocks the exact action that produces the risk. UAT separated usefulness from trust: a system can surface valuable leads while users stay unsure of the reasoning, and a fluent draft can still need factual review, so evaluating a GenAI application means weighing task completion, ranking quality, factual support, user understanding, operational limits, and auditability together. GTM Agent now forms an integrated MVP whose safety case rests on evidence-constrained generation, human review, restricted tools, offline-by-default agent behaviour, and traceable public data; the next stage should keep testing those boundaries, especially email-claim verification, operator access control, and broader user and fairness evaluation.

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Appendix A — Evaluation Protocol (UAT Script)

Supporting material for Part III (User Acceptance Testing Report). Each participant was assessed against the identical script and rating instrument.

Background. How many years have you worked with tenders?

Scenario 1 — Lead Discovery & Justification. *Task:* find a tender in your sector and identify the three most promising companies to approach. *Observed:* can they find the tender list, understand the rankings and score explanations, and do they need help? *Rating:* how plausible are these companies as bidders for this tender? (1–5)

Scenario 2 — Weight Adjustment. *Task:* find the weighting controls, adjust the ranking, review the new results, and explain whether the ranking changed as expected. *Observed:* can they discover the controls themselves and understand why the ranking changed? *Rating:* how much do you trust the ranking explanations? (1–5)

Scenario 3 — Email Generation & Verification. *Task:* generate an outreach email for one company and review it before sending, checking supporting evidence. *Observed:* can they select a company, generate, and read the email; do they understand where the evidence comes from; how much editing is required? *Rating:* rate the generated email (1–5) — could you send it after only minor edits?

Overall survey. How useful would this tool be for your work, and how much do you trust the matching agent? (1–5) How likely are you to recommend it to a colleague? (0–10)

Open feedback. What do you like most? What feature is missing? If you could change one thing, what would it be? Would you actually use this tool, and why?

Rating scales. All per-scenario and agent-trust items use a 1–5 Likert scale (1 = strongly negative . . . 5 = strongly positive). The final recommendation item uses a 0–10 (NPS-style) scale.

Acknowledgment of GenAI Usage

All four documents in this submission were written with the help of AI-based writing assistants — principally a Claude-based assistant, with OpenAI ChatGPT additionally used while preparing the User Acceptance Testing report — using the team's own project materials (the code base, working notes, and earlier drafts), and were then reviewed, corrected, and finalised by the team.

Concretely, GenAI was used to:

- draft and refine wording for clarity and readability;
- organise each report and produce its LaTeX layout; and
- help generate supporting visualisations, tables, and diagrams from our own data.

It was used as an assistant, not as an unchecked author: every factual claim, number, and figure was verified by the team against the project's own sources, and the content was edited to meet the quality bar expected of the submission. Within the product itself, GTM Agent uses Anthropic Claude for the orchestration agent and evidence-grounded email drafting, and open-source models for the matching pipeline.